GymPassJUST discussion summary

SE440 CSP11 | Version 1.2.2 | 2 November 2026

1. What does the project do?

It manages gym plans, class bookings and attendance. A student buys a plan and books a class. The instructor publishes a class and scans student QR codes. The manager handles accounts, plans and reports. Pages switch between Arabic and English.

2. How is attendance recorded?

The instructor chooses their own running class and scans the student's QR. The server checks the code, membership and booking. Both an active suitable plan and a confirmed booking are required. If either is missing, it returns an error. Otherwise it saves attendance. A repeated scan does not create another record.

Scanning works from the beginning to the end of the class. A booked student without a scan is absent after the end. QR codes last two minutes and refresh every 30 seconds. There is no manual student attendance button or separate security/reception account.

3. What are the other rules?

Booking opens 48 hours before class. Student booking cancellation requires at least two hours. A full class has a FIFO waiting list: the first eligible waiter gets a cancelled seat. The assigned instructor or manager may cancel a class until it ends. Cancelled and completed classes leave the timetable after 24 hours; history stays saved. The same active or queued plan cannot be bought twice.

4. How does the code work?

HTML and CSS display the page. Browser JavaScript sends a request. src/server.js receives it. src/service.js checks the gym rules and uses SQLite. The server sends a result back to the page. Passwords are saved as hashes, and private cookies identify signed-in users.

5. What do the planning numbers mean?

Four students x eight hours x 14 weeks = 448 hours. This includes 384 planned hours and 64 risk hours. At an assumed JOD 5/hour, the baseline is JOD 2,240. An extra JOD 224 sponsor reserve gives a JOD 2,464 ceiling. These are estimates.

The NPV example uses JOD 1,000 net annual benefit for three years at 8% and gives about JOD 337.10. The EVM example uses PV=800, EV=720 and AC=900: SPI=0.90 and CPI=0.80. It illustrates delay and higher cost; it is not a record of actual team spending.

6. What was tested?

The 30 existing tests pass. Statement coverage is 93.92%. The local 300-request load check has 20 confirmed and 280 waiting for 20 seats, with p95 894.84 ms. Real camera devices, independent user acceptance and university hosting remain pending.

Suggested explanation for four speakers

1. Explain the problem, users, schedule and cost estimate.

2. Show student signup, a plan, a booking and the language switch.

3. Show the instructor class and QR result, then explain the server checks.

4. Explain the tests, backup, remaining work and startup steps.

This is a rehearsal plan. The team must record its actual contributions separately. AI assistance is declared in the main report appendix.